Research on E-commerce Customer Segmentation Strategy Based on K-Means Clustering Algorithm

L H ZHANG
School of Computer Science
Nanjing University of Posts and Telecommunications
Jiangsu,Nanjing,China
1025040941@njupt.edu.cn

Abstract*—In the highly competitive e-commerce landscape, understanding customer behavioral patterns is critical for enterprises to implement precision marketing and optimize resource allocation. However, identifying high-value customers from massive behavioral data remains a complex challenge. This paper presents a customer segmentation approach using the K-Means clustering algorithm. To robustly evaluate the algorithm's performance in a controlled environment, we utilized a synthetic dataset simulating e-commerce user behaviors, focusing on two key dimensions: Average Session Duration and Monthly Spending. The optimal number of clusters (k=4) was determined by cross-referencing the Elbow Method and Silhouette Analysis. The experimental results successfully identified four distinct customer segments: "Core Loyal Users" (High Duration, High Spending), "Goal-Oriented Buyers" (Low Duration, High Spending), "Price-Sensitive Browsers" (High Duration, Low Spending), and "Low-Value Users". These findings demonstrate that K-Means is an effective tool for visualizing customer value hierarchies, enabling managers to formulate differentiated strategies such as immersive experiences for loyal users and efficiency optimization for goal-oriented buyers.

***Keywords*—Big Data Analytics, K-Means Clustering, Customer Segmentation, E-commerce, Precision Marketing**

I. INTRODUCTION

With the rapid advancement of the digital economy, E-commerce has evolved from a simple transactional channel into a complex ecosystem driven by user experience and data intelligence. In this highly competitive landscape, the cost of acquiring new traffic is skyrocketing, forcing enterprises to shift their strategic focus from "extensive growth" to "intensive management" of existing customers. During user interactions with platforms, massive amounts of behavioral data are generated, encompassing browsing history, click-through rates, transaction records, and session duration. These data possess the typical "4V" characteristics of Big Data (Volume, Velocity, Variety, and Value) and contain inherent commercial logic, reflecting consumer purchasing power and engagement levels. Consequently, Customer Relationship Management (CRM) based on big data analytics has become a critical engine for sustaining business growth. Despite the abundance of user behavioral data, extracting actionable insights from these vast records remains a significant challenge for e-commerce managers. Traditional analysis methods often rely on single-dimensional metrics (e.g., total sales volume) or simple statistical reports, which fail to capture the heterogeneity of user behavior. For instance, a "high-spending" user might be an efficiency-driven buyer who spends little time browsing, while a user with "long session duration" might be a price-sensitive "window shopper" with low conversion rates. A "one-size-fits-all" marketing strategy ignores these nuances, leading to resource wastage and low marketing ROI. Therefore, utilizing machine learning algorithms to automate the pattern recognition of Session Duration and Spending Power has become an imperative need to achieve Precision Marketing. To address the aforementioned challenges, this paper aims to construct a customer segmentation model employing the K-Means Clustering Algorithm to perform Unsupervised Learning on e-commerce user behavior. Instead of using noisy and potentially privacy-sensitive real-world data, this study utilizes a controlled Synthetic E-commerce Dataset generated via Gaussian distribution simulation, comprising 3,000 instances. This approach allows for a rigorous validation of the algorithm's mathematical convergence and clustering performance. This paper selects "Average Session Duration" (Time) and "Monthly Spending" (Money) as the core dimensions for dimensionality reduction and analysis. The objective is to algorithmically partition users into distinct value clusters. By characterizing these groups, this study seeks to provide enterprises with data-driven Customer Segmentation strategies, thereby maximizing operational efficiency and user retention.

II. RELATED WORKS

A. Basic Definition and Concepts of Cluster Analysis

Let the dataset be denoted as T. According to the features of the data, it is divided into x classes, namely $X_1, X_2, \ldots, X_x$. The process of clustering the dataset can be represented as follows: for every data sample or data object X_i in the sample space T (where $i=1,2,\ldots,m$), through different clustering algorithms, we place similar samples or samples with common characteristics into a certain region m, and X_i falls into only one region, satisfying the conditions of formula 1 below:

$$\begin{cases} T_1 \cup T_2 \cup T_3 \ldots \cup T_x = T \\ T_i \cap T_j = \emptyset \end{cases} \quad (1)$$

Through cluster analysis of the dataset, we can divide the total sample data space into several classes based on certain characteristics, user requirements, similar features, and other conditions. Thus, the classification of sample data is achieved in an unsupervised manner, which is of great help to subsequent data analysis, user personalized recommendation, and other applications .We can optimize the clustering algorithm based on the number of iterations, the selection of clustering centers, the initialization time of clustering centers, etc. Research by domestic and foreign scholars on various clustering algorithms usually focuses on the following aspects: how to reduce clustering time and how to enhance algorithm robustness.

B. The K-means Algorithm

The K-means algorithm is currently a very common method of cluster analysis. The prominent advantages of this algorithm lie in its simple principles and highly efficient application, making it very suitable for processing large-scale data. It has achieved good application results in many fields, including data analysis, personalized recommendation, data classification, and image recognition.

The K-means algorithm is a clustering algorithm with significant influence in the field of data mining. In many fields, such as schools, there is a massive amount of data information regarding the daily consumption, money transfers, and course schedules of different student groups. The K-means algorithm can be used to process this large amount of data, facilitating the school's effective management of various aspects of the students.

The initial cluster center is one of the important concepts in the K-means algorithm and often affects the clustering results. When initial cluster centers are used for clustering, the center points of the clusters are determined from the chaotically distributed dataset through the first round of algorithm iteration. Through continuous cyclical iterations, the cluster centers are constantly updated, causing the large volume of diverse data in the dataset to be distributed according to different categories.

For the traditional K-means algorithm, whether the retrieval results are accurate and the retrieval process is efficient is largely determined by the initial cluster centers. That is, the algorithm is relatively sensitive to the initial cluster centers, so appropriate initial centers must be selected; different initial cluster centers will lead to significantly different clustering results. If there are outliers in the data, the calculation of the initial cluster centers will be affected, leading to inaccurate clustering results.

To address the above problems, the K-means++ algorithm optimizes the selection of initial cluster centers [1]. The K-means++ algorithm selects initial cluster centers by calculating the shortest distance between each sample and the currently existing cluster centers. Although it improves clustering accuracy, the performance is still not ideal.

As the traditional K-means algorithm exhibits increasing limitations in practical applications, experts and scholars in relevant fields have attempted to improve and optimize it. For instance, some scholars constructed a density-based improved K-means algorithm[2], characterized mainly by its ability to overcome the issue of high sensitivity to initial centers. Others improved the traditional algorithm by combining density with the nearest neighbor method[3]. Some researchers improved the algorithm to meet the specific needs of text feature selection[4], effectively reducing the adverse impact of initial center randomness on the results. Additionally, scholars have established a K-means clustering algorithm using information entropy for precise attribute weighting[5]. Furthermore, addressing the practical scenario of plant leaf recognition, some scholars established an improved K-means algorithm based on cosine similarity[6].

III. Methodology

A. Data Acquisition and Description

Unlike traditional studies that rely on scraped social media data, which often suffer from noise, missing values, and privacy concerns, this study utilizes a Synthetic E-commerce Dataset. The data was generated using the Gaussian blob generation method (make_blobs) to simulate a controlled environment for rigorous algorithmic validation.The dataset consists of 3,000 instances, representing individual user sessions. It focuses on two core behavioral dimensions that are critical for customer segmentation:

- Session Duration (Minutes): Reflecting the user's engagement time on the platform.
- Monthly Spending (RMB): Reflecting the user's purchasing power and monetization value.

B. Data Preprocessing

1) Raw data generated from a Gaussian distribution may contain negative values or decimal figures that do not align with real-world business logic. To address this, we implemented a three-step preprocessing pipeline:Min-Max Normalization (Feature Scaling): We applied Min-Max scaling to map the generated values to realistic business ranges.

- Session Duration was mapped to the range of [1, 90] minutes.
- Monthly Spending was mapped to the range of [100, 5,000] RMB.

2) Integer Casting: To simulate discrete log records, all floating-point numbers were rounded and cast to integers.

3) Standardization (Z-Score Normalization): Since K-Means is a distance-based algorithm, it is highly sensitive to the scale of features. The magnitude of "Spending" (up to 5,000) is significantly larger than "Duration" (up to 90). Without standardization, the "Spending" feature would dominate the distance calculation. Therefore, we applied StandardScaler to transform the data such that it has a mean of 0 and a standard deviation of 1.

C. K-Means Clustering Algorithm

The K-means algorithm, also known as the K-average algorithm, is a representative prototype of Partitioning Methods. Its fundamental principle is as follows:

Given a dataset with n known data objects, the algorithm employs an objective function minimization approach. It randomly selects data objects from the sample set to serve as initial cluster centers, where the number of selected objects is less than or equal to the total sample size. Each initial object represents a cluster, and the mean value of the data samples within is typically regarded as the center of the cluster.

For the remaining unselected objects, each is assigned to the nearest cluster based on the calculated distance between the object and each cluster center. Subsequently, the mean of each cluster is recalculated. This process repeats until the criterion function converges and the algorithm terminates. Due to its favorable statistical and geometric properties, K-means is extensively applied in the analysis of numerical data. Consequently, achieving the ultimate convergence of the criterion function is the primary objective of the K-means algorithm.

Let E denote the criterion function, defined as follows:

$$E = \sum_{i=1}^{k} \sum_{x \in C_i} | x - \bar{x} |^2 \tag{2}$$

E refers to the sum of distances between all data objects and the cluster centers in the current dataset. A larger E value indicates greater distances between data objects and cluster centers, implying lower similarity among data objects within the cluster. Conversely, a smaller E value indicates higher similarity among the data objects within each cluster.

Where x represents a data object in the cluster; $\bar{x}$ is the cluster center of cluster c_i; k is the number of clusters; and c_i denotes the i-th cluster in the set.

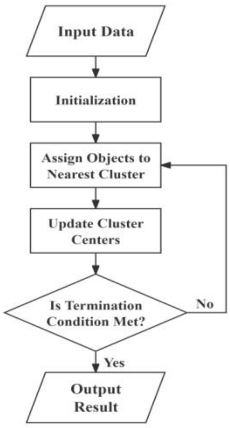

Fig. 1. K-Means Algorithm flowchart

The basic flow of the K-means algorithm is shown in Fig.1. For a sample set containing N objects, the specific steps of the K-means algorithm are as follows:

Input: A sample set containing N objects

Output: K clusters

1) Initialization: Determine the number of clusters, K, and determine the initial cluster centers (randomly select K objects).

2) Sample assignment: Based on the selected cluster centers, calculate the distance from each object to all cluster centers respectively, find the nearest cluster, and assign the object to that cluster.

3) Update cluster centers: Based on the latest sample assignment, recalculate the cluster center for each cluster.

4) Repeat steps *(2)* and *3)* until the cluster centers no longer change, and the algorithm terminates.

It can be seen from the steps of the K-means algorithm that the algorithm employs an iterative method, adjusting the assignment of objects within clusters each time and updating the positions of cluster centers based on the latest sample assignment. In the object assignment phase, similarity between objects and cluster centers is calculated mainly based on distance, generally using Euclidean distance.

D. K-Value Selection Strategy and Evaluation Metrics

One of the fundamental challenges in K-Means clustering is that the number of clusters,k, is a hyperparameter that must be pre-specified before running the algorithm. Choosing an inappropriate k can lead to suboptimal segmentation results, either by over-fragmenting coherent customer groups or by merging distinct groups into one.To determine the most appropriate k for our e-commerce dataset scientifically, rather than relying on arbitrary intuition, this study employed a cross-validation approach combining two widely accepted techniques: the Elbow Method and Silhouette Analysis.

1) The Elbow Method (Based on Inertia):The Elbow Method evaluates the model based on the Sum of Squared Errors (SSE), also known as inertia, which measures the internal compactness of clusters. Generally, as k increases, SSE naturally decreases because data points become closer to their respective centroids. The goal of this method is not to minimize SSE to zero (which occurs when k equals the total number of samples), but to identify the "elbow point"—the inflection point on the plot of SSE versus k where the rate of decrease sharply diminishes. This point typically represents the optimal trade-off between cluster compactness and model complexity.

2) Silhouette Analysis (Based on Structure):While the Elbow Method focuses solely on internal cohesion, Silhouette Analysis provides a more comprehensive evaluation by measuring both cluster cohesion (how close points are within a cluster) and separation (how far points are from neighboring clusters). The silhouette coefficient ranges from -1 to +1. A score closer to +1 indicates that sample points are well-matched to their own cluster and poorly matched to neighboring clusters, signifying a high-quality clustering structure.

IV. Experimental Results and Analysis

In this chapter, we present the empirical results obtained from applying the K-Means clustering algorithm to the synthetic e-commerce dataset. The analysis is divided into three parts: the determination of the optimal number of clusters, the visualization of the clustering results, and the detailed interpretation of each customer segment.

A. Determination of Optimal K

To ensure the validity of the segmentation, we employed both the Elbow Method and Silhouette Analysis to select the optimal number of clusters, k .Figure 2 illustrates the comparative results of these two metrics for k ranging from 2 to 9.

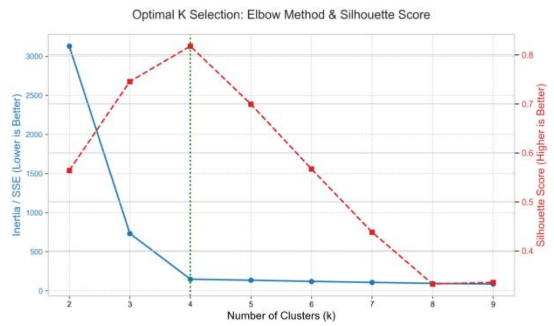

Fig. 2. Optimal K selection using Elbow Method (Blue, Left Axis) and Silhouette Score (Red, Right Axis)

As shown in Fig.2, the blue line representing the Sum of Squared Errors (SSE) exhibits a sharp decrease from k=2 to k=4. Beyond k=4, the rate of descent significantly flattens, forming a distinct "elbow" point. Simultaneously, the red dashed line representing the average Silhouette Score reaches a relatively high and stable value around k=4 and k=5. Although k=5 yields a slightly higher silhouette score, the marginal gain in cluster quality does not justify the added complexity over k=4, which offers a clearer structural break in the Elbow plot.Therefore, based on the comprehensive assessment of both metrics, we determined k=4 as the optimal number of clusters for subsequent analysis.

B. Clustering Results Visualization

With the optimal k=4, we applied the K-Means algorithm to the standardized dataset. To intuitively visualize the segmentation structure, we plotted the results on a 2D scatter plot using the original (unscaled) features: "Session Duration" and "Monthly Spending"

Fig. 3. Visualization of K-Means clustering results with k=4

The x-axis represents average session duration (min), and the y-axis represents monthly spending (RMB). The yellow stars denote cluster centroids.Figure 3 demonstrates that the algorithm successfully partitioned the 2,000 users into four distinct and well-separated clusters. The boundaries between clusters are relatively clear, indicating a high quality of segmentation. The four centroids (marked by yellow stars) effectively represent the "average user profile" within each segment, located at different quadrants of the duration-spending plane.

C. Cluster Profiling and Interpretation

To understand the business value of each segment, we analyzed the statistical characteristics of the four clusters. Table 1 summarizes the average "Session Duration" and "Monthly Spending" for each cluster center.

TABLE I. STATISTICAL PROFILES OF THE FOUR IDENTIFIED CUSTOMER CLUSTERS

Cluster Label	Avg.Session Duration/Minutes	Avg.Monthly Spending/RMB
0	22.3	663.4
1	43.5	4234.3
2	77.3	2660.9
3	13.0	3840.4

Based on the data presented in Table 1, we provide an in-depth interpretation of the behavioral patterns and business value for each of the four identified segments in numerical order.

1) Cluster 0: Low-Value Users (Low Duration, Lowest Spending)

- Characteristic Analysis: This group scores lowest on both dimensions. The average session duration is 22.3 minutes, and the average spending is only 663.4 RMB, which is significantly lower than all other groups.
- Business Value: They represent the platform's "long-tail" or dormant users. Their current contribution and activity levels are minimal, and they carry the highest risk of churn.

2) Cluster 1: Core Loyal Users (Medium-High Duration, Highest Spending)

- Characteristic Analysis: This group constitutes the platform's "absolute consumption powerhouse." Their average monthly spending reaches a substantial 4234.3 RMB, the highest among all clusters. Simultaneously, their average session duration is 43.5 minutes, representing a healthy level of engagement. They achieve an optimal balance between time investment and monetary output.
- Business Value: They are the current primary revenue pillar and core assets of the platform, holding the highest retention priority.

3) Cluster 2: Price-Sensitive Browsers (Highest Duration, Moderate Spending)

- Characteristic Analysis: They contribute the longest average session duration, reaching 77.3 minutes, indicating extremely high stickiness. However, their average spending level of 2660.9 RMB is moderate compared to the high-spending groups. This suggests they spend significant time "browsing," comparing prices, or waiting for promotions, typical of "Window Shoppers".
- Business Value: They are the primary contributors to platform activity and traffic volume, holding vast potential for conversion, though their current purchase decision cycles are longer.

4) Cluster 3: Goal-Oriented Buyers (Lowest Duration, Extremely High Spending)

- Characteristic Analysis: This is a highly interesting and high-value segment. Their average session duration is the shortest among all groups, at only 13.0 minutes, yet their average spending is very high at 3840.4 RMB. This extreme "low-time, high-output" characteristic suggests highly purposeful shopping behavior. They are typical efficiency-driven users who "buy and go".
- Business Value: They are "efficiency-driven cash cows" with extremely high conversion efficiency.

D. Comparative Analysis:DBSCAN

To validate the intrinsic stability of the customer segments identified by K-Means, this study employs the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm as a comparative benchmark. While K-Means relies on Euclidean distance minimization and assumes convex, spherical clusters, DBSCAN defines clusters as continuous regions of high point density, separated by regions of low density. This fundamental difference allows us to verify whether the four-cluster structure is a genuine topological property of the dataset or an artifact of the K-Means objective function.

1) Determination of Optimal ϵ

The performance of DBSCAN is sensitive to two parameters: the scanning radius (ϵ) and the minimum number of points required to form a dense region ($min\_samples$). Instead of arbitrary selection, we adopted the sorted k-distance graph heuristic proposed by Ester et al. to mathematically determine the optimal ϵ .

The underlying theory posits that in a cluster-bearing dataset, points within a cluster will have small k-nearest neighbor distances, while noise points will exhibit significantly larger distances. By plotting the distance to the k-th nearest neighbor (k is set to $min\_samples$ = 10) for all data points in ascending order, we can observe the density distribution of the dataset.

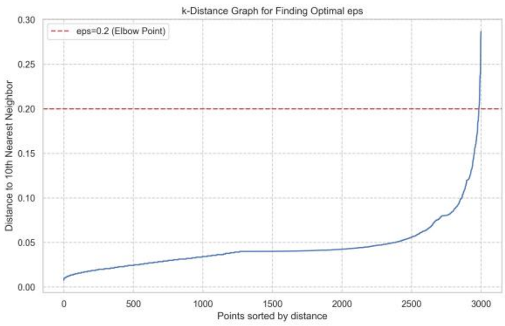

Fig. 4. Sorted k-distance graph for optimal ϵ

As presented in Fig. 4, the curve remains initially flat, representing points within high-density clusters. A distinct "elbow" or "knee" point emerges at approximately 0.2 on the y-axis. This inflection point represents a critical phase transition in density, where the distance required to connect additional points increases drastically. Consequently, ϵ=0.2 was selected as the optimal threshold to maximize cluster cohesion while effectively filtering background noise.

2) Visualization of DBSCAN Results

Applying the derived parameters (ϵ=0.2,$min\_samples$=10), the DBSCAN algorithm converged to a solution that independently corroborates the K-Means results.

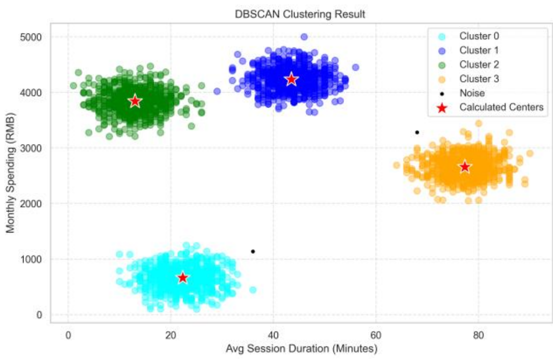

Fig. 5. Visualization of DBSCAN clustering results with ϵ=0.2

The results, visualized in Fig. 5, demonstrate two significant theoretical insights:

- Intrinsic Cluster Validity: Without any prior knowledge of k, DBSCAN successfully resolved the data into four disjoint dense sets. This automatic convergence serves as strong evidence that the "four-segment structure" is inherent to the user behavior distribution and not a forced partition.
- Noise Isolation Capability: Unlike K-Means, which forces every point into a cluster (potentially distorting centroids), DBSCAN identified a subset of points as Noise (represented by black dots in Fig. 5). These points represent users with erratic behavioral patterns (e.g., extremely high spending with zero stickiness) that violate the Gaussian assumptions of the main clusters. The exclusion of these outliers suggests that the four identified core segments are structurally robust and statistically significant.

V. Conclusion and Future Work

A. Conclusion

This study aimed to explore the application of the K-Means clustering algorithm in e-commerce customer segmentation. By constructing a controllable synthetic dataset based on Gaussian distributions, we focused on two critical behavioral dimensions: "Session Duration" (User Stickiness) and "Monthly Spending" (Monetary Value).

The primary contributions and findings of this research are summarized as follows:

*1) Methodological Rigor:*We successfully established a standardized workflow for segmentation analysis, moving from synthetic data generation and rigorous preprocessing (including Min-Max scaling and standardization) to scientific model selection. The optimal number of clusters, k=4, was determined through a combined assessment of the Elbow Method and Silhouette Analysis, ensuring the statistical validity of the segmentation.

*2) Identification of Distinct Segments:*The K-Means algorithm successfully partitioned the user base into four behaviorally distinct segments with clear boundaries. This demonstrates the algorithm's effectiveness in identifying underlying patterns even within a limited two-dimensional feature space.

*3) Actionable Business Insights:*Based on the empirical results, we defined four highly actionable customer profiles:

- Core Loyal Users (Cluster 1): The platform's primary revenue engine, characterized by high engagement and high spending.
- Goal-Oriented Buyers (Cluster 3): Highly efficient "cash cows" with short session times but substantial contributions.
- Price-Sensitive Browsers (Cluster 2): High-traffic contributors who exhibit high stickiness but low immediate monetization.
- Low-Value Users (Cluster 0): The baseline group with minimal activity and high churn risk.In conclusion, this study provides a clear theoretical framework and practical demonstration of how data-driven techniques can transform raw behavioral data into strategic marketing insights.

B. Limitations

While this study offers valuable insights, several limitations must be acknowledged, which stem primarily from the experimental design:

*1) Synthetic Data Constraint:*The most significant limitation is the reliance on synthetic data generated via make_blobs. While this provided ground truth for model validation, the data distribution is ideally Gaussian and lacks the noise, outliers, complex correlations, and sparsity inherent in real-world e-commerce logs. Therefore, the model's robustness on messy, real-world data remains untested.

*2) Limited Feature Dimensionality:*The study focused solely on "Duration" and "Spending." It omits other critical dimensions of the classic RFM model, such as "Frequency" (how often users buy). Furthermore, it does not account for categorical data like product preferences, demographics, or geographic location, which would provide a more holistic user view.

*3) Algorithm Specificity:*K-Means assumes clusters are spherical and equally sized. While appropriate for our Gaussian-generated data, it might struggle with clusters of arbitrary shapes or varying densities often found in complex real-world scenarios.

C. Future Work

To address the aforementioned limitations and deepen the research, future work could focus on the following directions:

1) Validation on Real-World Data: The primary next step is to apply the established pipeline to a large-scale, real-world e-commerce dataset. This would allow for an assessment of the model's performance under noisy conditions and provide more practical business case studies.

2) Feature Expansion: Future models should incorporate additional features. Adding "Purchase Frequency" would complete the RFM framework. Including categorical features (e.g., product categories browsed) would require adopting algorithms capable of handling mixed data types, such as K-Prototypes.

3) Comparative Algorithm Analysis: It would be beneficial to compare the performance of K-Means with other clustering algorithms, such as Gaussian Mixture Models (GMM, which offers soft clustering probabilities), to determine the most suitable approach for different types of data distribution*s*.

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